Decarbonising today's power system

Expansion at European scale

These solutions are written to be read after the workshop, one subsection per question. The keys of the closed questions are given in the worksheet's lettering. Each open question has a model answer, which is one good answer among several, as in the Model Lab. Figures are ESM's own from section 8, or drawn from the Explorer's data, and each caption says which. Explorer views are named as on the worksheet, as tab › figure, and linked. Unless the text says otherwise, a scenario is at a budget of 1 %, a cut of 99 %. Where the evidence is a number the figure does not draw, press *Table* at the figure's top right, which lists every number behind it.

1 The baseline: prices and the last quarter

Part 1 · ESM §8.2

1. What the cut changes

(i) Two different cuts. The first three quarters of the cut and the last quarter are made in different ways, and the last costs far more (Figure S1). Up to a cut of 75 %, new solar and onshore wind, with eight-hour batteries and the existing biomass plants run harder, push coal and lignite out of the generation, and some of the natural gas. The system cost rises by about a third and shifts from operating to capital: fuel is saved, and wind, solar and batteries are paid for. The price is set far less often by plants burning fuel and far more often by batteries, wind and solar (Figure S2(b)), and more interconnectors are bought as the budget tightens.

What is left for the last quarter is gas burnt in the hours with little wind and sun, which more wind, solar and daily batteries barely reach. The last quarter still builds more of all three, but it is cut by a change of fuel and by machines for a few weeks: every gas plant switches to biomethane, a dear fuel, and new gas turbines are built to run in a few winter weeks (about 13.5 GW, some 700 hours a year). Capital and operating costs both rise, and the system cost rises by more over the last quarter than over the first three (from 40.5 to 55.8 bn€ a year, against about 31 to 40.5); the carbon price climbs from 66 to 286 €/t. In the Explorer: Overview › *Energy balance: all twelve zones* and Overview › *What is built: all twelve zones*, *Baseline, budget 1 %* or *Baseline, budget 25 %* against *Baseline, no budget*; the carbon price and the system cost are tiles of Overview › *Headline numbers*.

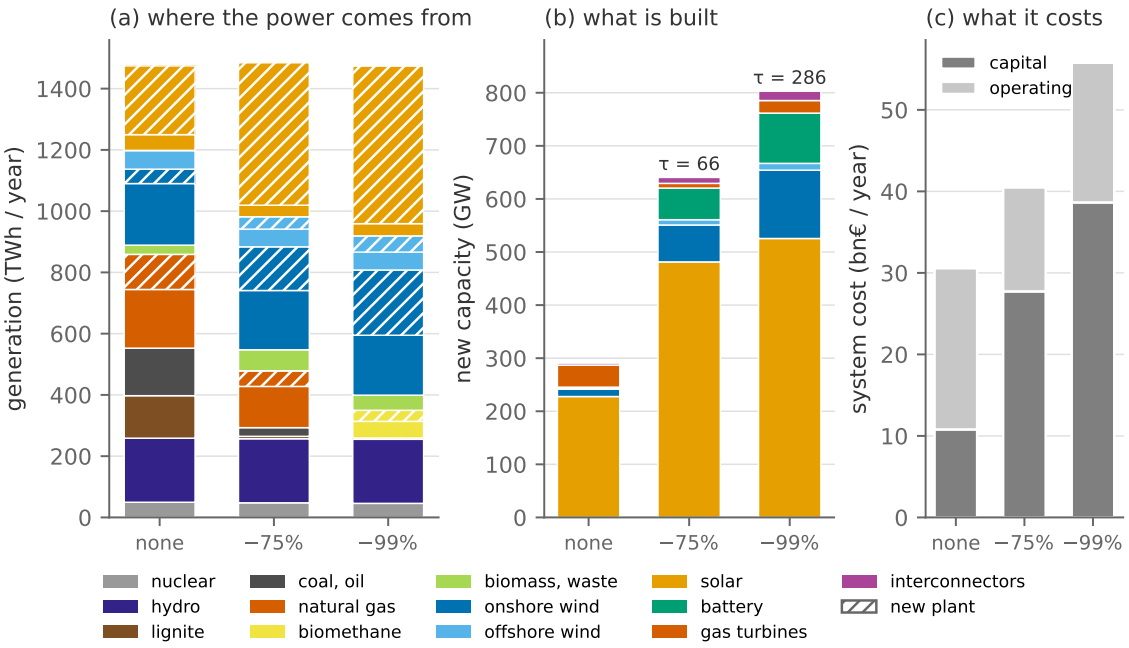


Figure S1: ESM, Fig. 8.1: the baseline without a budget and at cuts of 75 and 99 %: (a) generation by source, existing plants solid and new plants hatched; (b) new capacity by technology, interconnectors included, with the carbon tax in €/t above each bar; (c) the annual system cost, capital and operating.

(ii) Key: (c). The cut fattens both tails of the price duration curve (Figure S2(a)): more hours near zero (868 hours below 5 €/MWh, against 155 with no cut) and many more dear ones (2,109 above 100 €/MWh, against 119). Both ends are set by something other than storage. In the cheap hours wind and solar are curtailed and set the price at a running cost near zero: at a cut of 99 % they set it for about a sixth of the year's demand, against 3 % with no cut. In the dear hours a plant burning fuel sets it at its running cost, carbon included, and both fuels have become dear: coal pays 286 €/t for its carbon, and the gas plants burn biomethane (about 108 to 116 €/MWh in a combined cycle, depending on its efficiency, and about 150 €/MWh in an open-cycle turbine). Storage, the batteries and the hydro reservoirs, sets the price for almost two thirds of the year's demand, but it has no price of its own: it carries the two levels between hours, net of its losses. So the average price rises (44 to 68 €/MWh), although wind and solar cost nothing to run. In the Explorer: Prices and margins › *The price-duration curve* and Prices and margins › *What sets the price*, *Baseline, budget 1 %* against *Baseline, no budget*.

2. What covers the last quarter

(i) Long storage. Every curve of Figure S3(b) is at one at eight hours, the duration the model builds, where a MW earns exactly its annual cost (Result 6), and the deeper the cut, the higher the curve lies beyond it: storage that holds days or weeks is worth

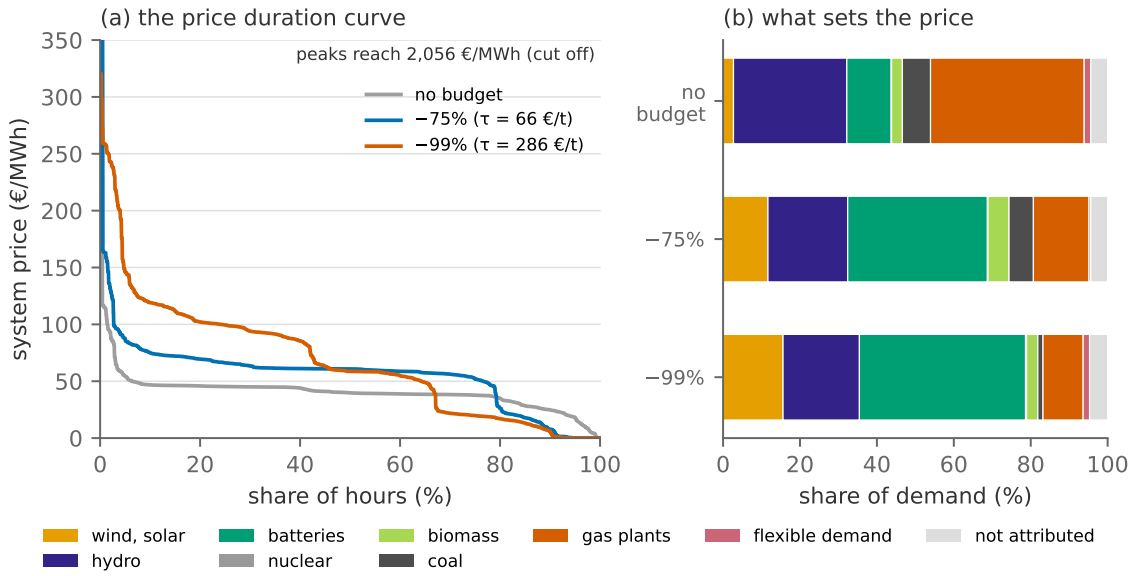


Figure S2: ESM, Fig. 8.2: the baseline without a budget and at cuts of 75 and 99 %: (a) the system's price duration curve, each hour's zonal prices weighted by the zones' demand; (b) what sets the price, as shares of the year's demand.

more, relative to its cost, the more the budget bites (a MW that holds a week would earn 10 % of its cost with no cut and 29 % at a cut of 99 %). What a MW earns in a year is the product of three factors,

$$\text{full cycles a year} \times \text{duration} \times \text{earnings per MWh delivered},$$

since its cycles times its duration are the MWh it delivers. Figure S4 draws the first and the last. The cycles fall with the duration, to about twenty a year for a week, and the cut barely moves them. What the cut changes is the margin on each MWh: with no cut a MWh delivered earns about the same whatever the duration, while at a cut of 99 % storage that holds days or weeks earns two to three times as much per MWh (about 90 against 34 € for a week). The budget brings long stretches of cheap hours, with wind and sun curtailed, and of dear ones, the weeks with little wind and sun when the gas plants burn biomethane. In Germany the price is near zero at midday from April to August and high in every hour from autumn to early spring, where with no cut it is nearly flat through the year (Operation › *The price through the day, month by month*, zone Germany, *Baseline, budget 1 %* and *Baseline, no budget*).

Still none of it pays. A MW that holds d hours costs $9.7 + 6.4\,d$ k€ a year, almost all of it energy capacity once d is long, so its cost grows in proportion to the duration, while what it earns grows far more slowly: a MW that holds a week costs about 1.1 M€ a year and would earn about 0.3 M€ at a cut of 99 %. Storage that holds a week would pay only with energy capacity at under a third of a battery's cost (28 %), the kind of cost that hydrogen in salt caverns has. And the dear prices it could sell into are capped:

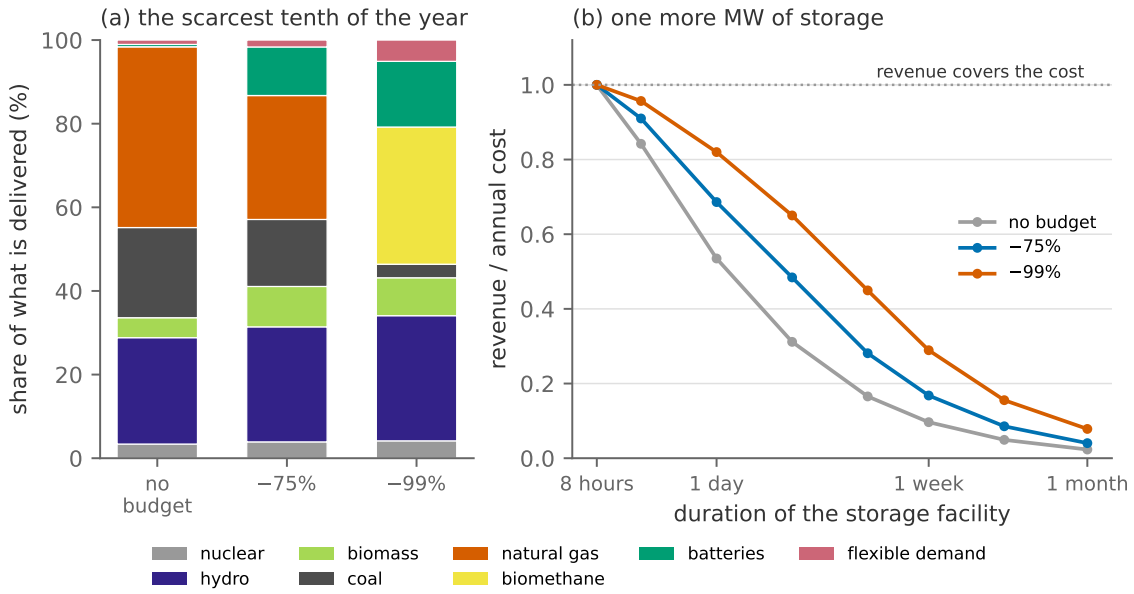


Figure S3: ESM, Fig. 8.3: the baseline without a budget and at cuts of 75 and 99 %: (a) what covers the tenth of the year with the highest residual demand, as shares of what is delivered in it; (b) what one more MW of storage of each duration would earn trading against the solved prices, over its annual cost on the battery menu, in the zone where it earns the most.

biomethane, bought at one price all year, is seasonal storage that costs nothing to hold. In the Explorer: Prices and margins › *How often storage would cycle, and what each MWh would earn*; what a MW would earn and what it costs, duration by duration, are in the table of Prices and margins › *What one more MW of storage would earn* (press *Table*).

(ii) Key: (b). The block steps back when the price passes its willingness to pay, 100 €/MWh, so what matters is what sets the price in the hours gas used to cover. A new combined-cycle plant on natural gas costs $34.3 + 0.33\,\tau$ €/MWh to run, which passes 100 €/MWh at

$$\tau = \frac{100 - 34.3}{0.33} \approx 199 \text{ €/t},$$

and an open-cycle turbine ($46.8 + 0.46\,\tau$) passes it at about 116 €/t. At the 66 €/t of a cut of 75 %, every gas plant on natural gas runs for less than 100 €/MWh, and the price reaches the block only in the scarcest hours. At the 205 €/t of a cut of 90 %, even the best of them costs more to run than the block will pay, so in every hour in which a gas plant on natural gas would have set the price, the block steps back first. Nothing in the model limits how often it may do so, which ESM's bias box in section 8 counts as understating the carbon price at the deep end. In the Explorer: Sweeps, *Flexible demand steps back* against *CO2 budget, all twelve zones*; the carbon price at each budget is the tile "Carbon price" of Overview › *Headline numbers*.

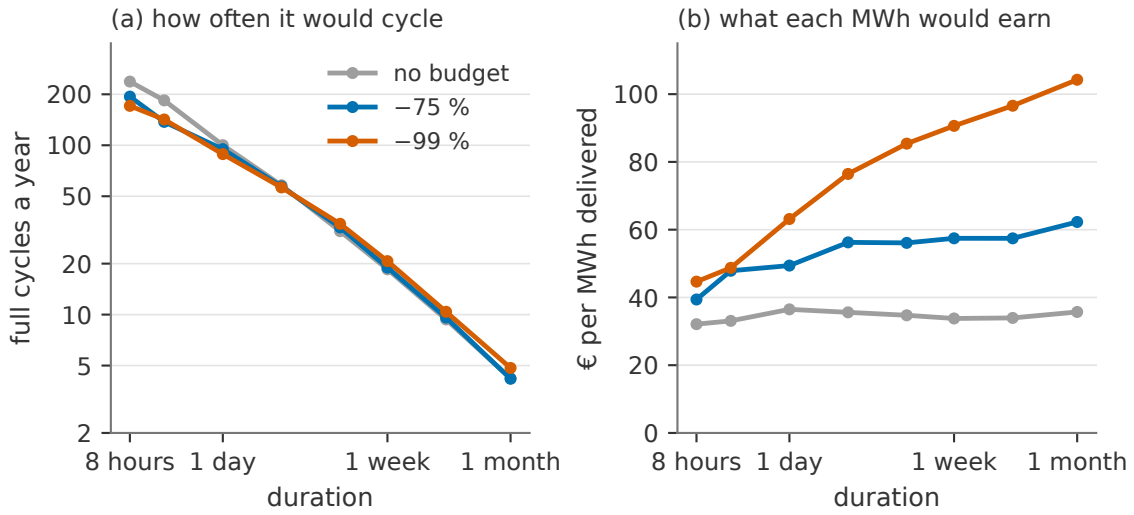


Figure S4: Drawn from the Explorer's data, as its Prices and margins › *How often storage would cycle, and what each MWh would earn* draws it, for the three budgets of Figure S3: (a) the full cycles a year that one more MW of storage of each duration would run, the energy it delivers over the energy it holds; (b) what it would earn per MWh it delivers, the price it sells at less what the energy cost it, losses and operating charge included. Each duration in the zone where it would earn the most over its cost.

Stretch question

(i) Result 7. Differentiating the interconnector's terms of the Lagrangian (ESM §8.2) with respect to its flow in one hour gives

$$f_{k,h}: \quad \eta_k \lambda_{b_k,h} - \lambda_{a_k,h} + \underline{\omega}_{k,h} - \overline{\omega}_{k,h} = 0,$$

with complementary slackness $\underline{\omega}_{k,h}\, f_{k,h} = 0$ and $\overline{\omega}_{k,h}\,(\bar{f}_k - f_{k,h}) = 0$. Take the three cases in turn. If what a MWh fetches at the far end, net of losses, is above its price at home, then $\overline{\omega}_{k,h} > \underline{\omega}_{k,h} \geq 0$: the cable runs full, so it is not at zero, $\underline{\omega}_{k,h} = 0$ and $\overline{\omega}_{k,h} = \eta_k \lambda_{b_k,h} - \lambda_{a_k,h}$. If it is below, $\underline{\omega}_{k,h} > 0$: nothing flows, the limit is slack and $\overline{\omega}_{k,h} = 0$. At equality both are zero. In every hour, therefore,

$$\overline{\omega}_{k,h} = \max\left(\eta_k \lambda_{b_k,h} - \lambda_{a_k,h},\ 0\right).$$

The capacity $\bar{f}_k$ enters the Lagrangian through its annual fixed cost F_k and through every hour's limit $f_{k,h} \leq \bar{f}_k$, so its condition is

$$\bar{f}_k: \quad -F_k + \sum_h \overline{\omega}_{k,h} + \underline{\beta}_k - \overline{\beta}_k = 0.$$

Substituting the first result and rearranging gives Result 7. The sum is the congestion rent a MW of the cable earns in a year: the price difference between its two ends, net of losses, in the hours it runs full. It is a generator's condition (Result 6) with the price at home in place of the running cost.

(ii) The planner and the owner. Below its ceiling both bound multipliers are zero, so one more MW of the cable earns exactly its annual fixed cost: the price difference between its two ends, net of losses, in the hours it runs full. That is also what it saves the system, since in each of those hours it lets one more MWh be bought where it is cheap and delivered where it is dear. The rent per MW falls as the cable grows, because more capacity narrows the spread it earns on, and the planner stops where it has fallen to the cost. An owner paid only the rent earns the rent per MW on its whole capacity. A MW more narrows the spread on every MW it already owns, so the last MW adds less to its income than that MW's rent, and it stops while the rent is still above the cost: less cable, and a wider spread, than the system should have. Section 6 measured both sides for the Skagerrak cables: the system's saving rose with their size, while the rent the cables collected peaked a little below today's size and then fell. Hence transmission that is regulated or planned jointly rather than left to merchant cables.

(iii) Key: (d). The condition compares what one more MW would earn on a border with what it costs there, not with what other borders earn. Between Germany and southern Norway the prices differ in most hours, so a MW more would earn about 290 k€ a year, but the cable is long and under the sea, about 4 M€ per MW or 380 k€ a year: the condition holds at its lower bound, $\underline{\beta}_k > 0$. At a cut of 99 % all three cases of Result 7 are on Germany's borders:

Border with Germany	a MW, k€ a year		Bought
	rent	cost	
western Denmark	228	228	970 of 2,000 MW: interior
eastern Denmark	186	74	to its ceiling: $\overline{\beta}_k \approx 110$ k€
southern Norway	290	380	none: $\underline{\beta}_k \approx 90$ k€

The rent is what the cable saves the system, so the planner counts it in full, whoever collects it. In the Explorer, Grid, emissions and money › *Interconnectors bought* shows what is bought on each border against its ceiling; no figure draws a cable's rent per MW, and the numbers above are from the solved model.

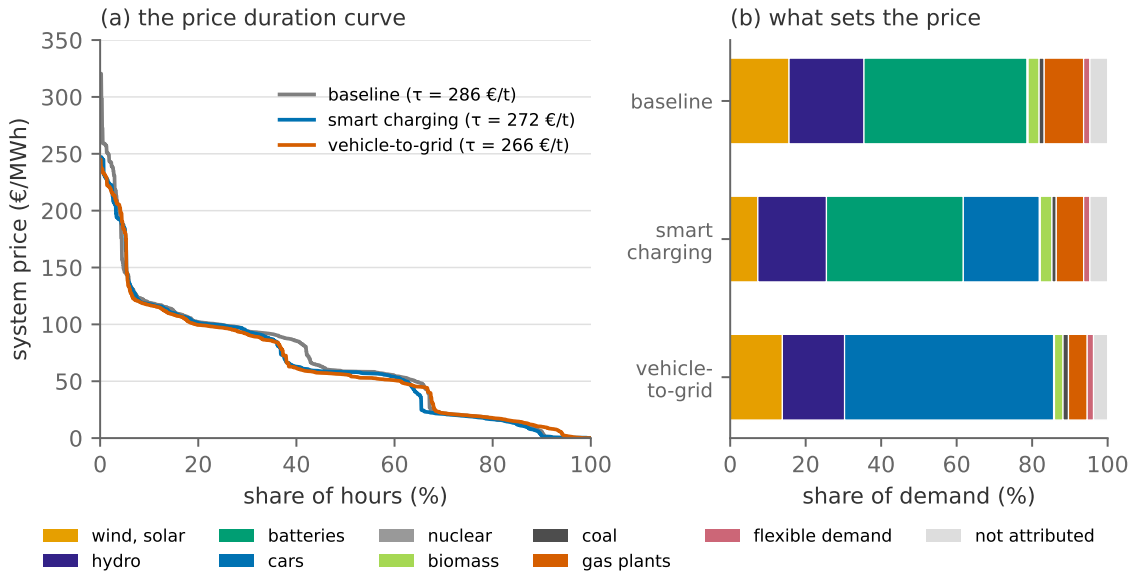


Figure S5: ESM, Fig. 8.4: the baseline, smart charging and vehicle-to-grid at a cut of 99 %: (a) the system's price duration curve; (b) what sets the price, as shares of the year's demand.

2 Electric vehicles: half a battery and a whole one

Part 2 · ESM §8.3

1. What the cars do to prices

(i) The two tails. Smart charging and vehicle-to-grid lower the dear tail about equally; they differ at the cheap tail, which only vehicle-to-grid lifts (Figure S5(a): the dearer half of the year averages about 109 €/MWh in the baseline, 104 with smart charging and 103 with vehicle-to-grid, while the hours below 5 €/MWh go from 868 to 930 with smart charging and down to 528 with vehicle-to-grid).

Sections 4 and 5 say what to expect. Demand that was already there and becomes flexible relieves both tails: it leaves the scarce hours and fills the cheap ones (ESM §5.2). A battery compresses the dear tail and leaves the cheap one where curtailed wind and solar set it (ESM §4.2). Smart charging makes the cars' existing demand flexible, and the dear tail falls as expected; the cheap tail stays, because the model builds fewer stationary batteries (79.7 against 94.7 GW). The cars take over the batteries' charging in the cheap hours instead of adding to it, and the batteries that remain still just pay for themselves (question 2). Vehicle-to-grid makes the cars storage that is large and costs the power system nothing: no stationary battery is built, and the cars buy enough in the cheap hours to lift them as well. What sets the price moves from the batteries to the cars (Figure S5(b)), wholly with vehicle-to-grid, and the average price falls a little either way

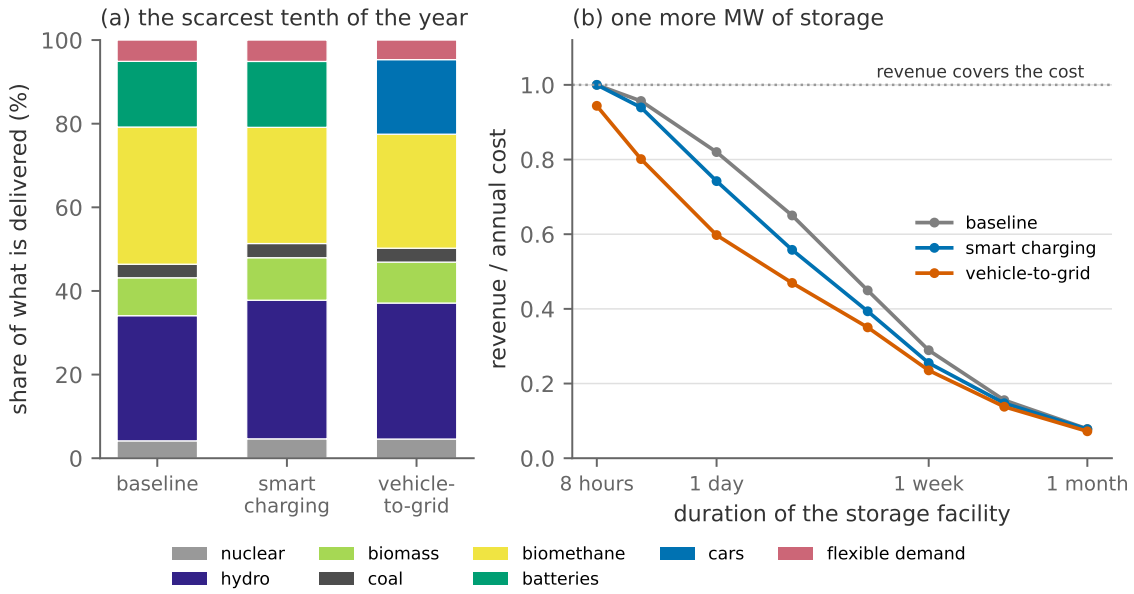


Figure S6: ESM, Fig. 8.5: the baseline, smart charging and vehicle-to-grid at a cut of 99 %: (a) what covers the tenth of the year with the highest residual demand; (b) what one more MW of storage of each duration would earn over its annual cost.

(68 to 64 and 65 €/MWh). In the Explorer: Prices and margins › *The price-duration curve*, *Vehicle-to-grid* against the *Baseline*, and *Smart charging* against it the same way; the stationary batteries built are the tile "Batteries" of Overview › *Headline numbers*.

(ii) Key: (a). The cars set the price the way a battery does (ESM, appendix E). Charging part-loaded, the cars' charging condition holds with both of its multipliers zero, so the price equals the value of what reaches their batteries, $\lambda_{z,h} = \eta^{EV}\theta^{EV}_{z,h}$; discharging part-loaded with vehicle-to-grid, it equals $\theta^{EV}_{z,h}/\eta^{EV}$, plus the small operating charge. Between the hours in which the cars' batteries reach their ceiling or their floor, $\theta^{EV}_{z,h}$ stays put, so the cars pass on the price of the hours they trade with, net of their losses: they have no price of their own. That their batteries cost the power system nothing explains why they take over from the stationary batteries, not what the price is when they set it. In the Explorer: Prices and margins › *What sets the price*, where the cars set the price for 55 % of the year's demand with vehicle-to-grid and 20 % with smart charging.

2. Cars against batteries

(i) The battery's investment condition. A stationary battery that is built below any limit earns exactly its annual fixed cost in the price swings it trades on (ESM §7.2):

$$\sum_h \left(\overline{\phi}^c_{s,h} + \overline{\phi}^d_{s,h}\right) + \bar{\delta}_s \sum_h \overline{\psi}_{s,h} = F_s,$$

and each multiplier on the left is a difference between the prices of different hours, net of losses. While batteries are built, this pins the daily spread at what pays a battery. The cars' storage is free, so with smart charging it takes over part of the arbitrage and the system builds fewer batteries, until those that remain again just pay: the cars replace batteries rather than narrow the spread, and the cheapest hours stay as they were. With vehicle-to-grid the cars offer so much free storage that no stationary battery is built. The condition then holds with $\underline{\beta}_s > 0$: a battery would earn less than its cost (94 % of it, Figure S6(b)), so nothing stops the spread from falling below what pays one. The cars buy in the cheapest hours and lift them: about 528 hours below 5 €/MWh, against 930 with smart charging (Figure S5(a)). In the Explorer: Prices and margins › *What one more MW of storage would earn*, *Vehicle-to-grid* against *Smart charging*.

(ii) Key: (c). The cars take the daily spread a battery lives on: with vehicle-to-grid they cover the batteries' share of the scarcest hours (Figure S6(a), about a sixth of what is delivered in them), and eight-hour storage loses its revenue to them. Storage that holds a week earns on the spread between a windy week and a calm one, and the cars cannot carry energy across a calm week: they offer half their batteries, drained by the driving whatever the price and kept above a floor for the morning's trips. Like the batteries they replace, they move energy within days, not across weeks, so a MW that holds a week earns almost what it does in the baseline (24 % of its cost, against 29 %). In the Explorer: Prices and margins › *What covers the scarcest hours* and Prices and margins › *What one more MW of storage would earn*, *Vehicle-to-grid* against the *Baseline*.

Beyond the question: a year in segments. The model solves the year in segments, which average away some of the swings a battery earns on. Solved hour by hour, vehicle-to-grid keeps about 38.5 GW of stationary batteries beside the cars, where the segments build none, and the cars give back less (ESM, appendix B and the bias box of section 8). The direction of part 2's results holds; how completely the cars displace the batteries is the part to hold loosely.

3 Hydrogen: a flexible load, not a storage

Part 3 · ESM §8.4 and §5

1. The hydrogen sector, part by part

(i) Part by part. Both sectors are a second market tied to electricity by a conversion (Table S1). The electrolyser plays the heat pump's role: it buys electricity and sells the other carrier, which links the two prices. It runs flat out when power is cheaper than the

Table S1: The hydrogen sector of ESM §8.4 read against the heat sector of ESM §5.

Hydrogen sector	Heat sector	The role
electrolyser	heat pump	buys electricity and sells the other carrier, which links the two prices
reformer, on natural gas or biomethane	gas boiler	the fuel-burning alternative, whose cost sets the carrier's price when it is at the margin
imports	a second boiler, in steps	an outside supply, the dearest step in use setting the price
hydrogen storage	thermal storage (discussed, not modelled)	separates when the carrier is made from when it is used
hydrogen turbine	none	a way back to electricity

hydrogen it makes, $\lambda_{z,h} < \eta^{EL}\lambda^{H_2}_{z,h}$, as the pump runs when $\lambda^E_h/COP_h < c^B$, and part-loaded it ties the two prices together, $\lambda_{z,h} = \eta^{EL}\lambda^{H_2}_{z,h}$, as the pump ties $\lambda^H_h = \lambda^E_h/COP_h$. The reformer plays the gas boiler's role: the fuel-burning alternative, which sets the price of hydrogen whenever it is at the margin, $\lambda^{H_2}_{z,h} = c_g + \tau e_g$, as the boiler sets $\lambda^H_h = c^B$, and with it what an electrolyser will pay for power. The imports are a second boiler that comes in steps, the dearest step in use setting the price. Hydrogen storage is the thermal storage that section 5 discusses and leaves out of its model: it separates when the carrier is made from when it is used, so the electrolysers can buy power when it is cheap although the demand for hydrogen is flat.

The comparison breaks down in three places. The turbine is a way back to electricity, which heat does not have. Hydrogen's demand is flat, where heat's is cold and peaky. And the electrolyser loses energy where the pump gains it ($\eta^{EL} = 0.7 < 1 < COP$, which averages 3.0). Besides, every part of the hydrogen chain is bought, where section 5's pump was given. In the Explorer: Hydrogen › *Hydrogen supply and use: all twelve zones*.

(ii) Key: (b), (d) and (f). A hydrogen turbine is the open-cycle gas turbine with another fuel, at the same cost, so it burns hydrogen only if hydrogen is the cheaper fuel in the hours when power is dear (ESM §8.4). Its rival is biomethane, bought in any month at one price (62 €/MWh): seasonal storage that costs nothing to hold. Holding hydrogen in a salt cavern from summer to winter costs more than that, about 89 € a year for each MWh of the cavern, before the electricity and the electrolyser that made the hydrogen. Even the hydrogen sector's own seasonal balancing is done by biomethane: the reformers burning it run hardest from autumn to early spring, when electrolysis makes the least, and almost no hydrogen storage is built (Figure S7(b)). So hydrogen can pay as seasonal storage only if biomethane goes, (b), or gets dear enough, (f); (d) is the condition itself.

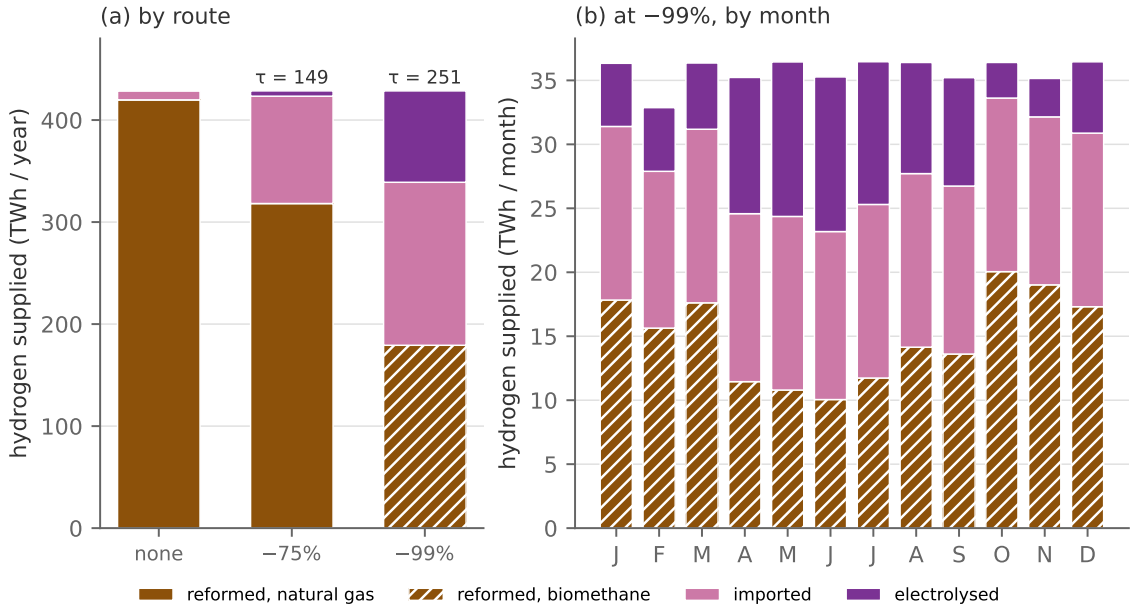


Figure S7: ESM, Fig. 8.7: (a) hydrogen supply by route (reformed from natural gas or from biomethane, imported, electrolysed) without a budget and at cuts of 75 and 99 %, with the carbon tax in €/t above each bar; (b) the same at a cut of 99 %, month by month, summed over the zones.

Without biomethane the model builds about 22 GW of hydrogen turbines and 38 TWh of hydrogen storage (Overview › *Headline numbers*, *No biomethane* against the reference). A much dearer biomethane works the same way, although the Lab has not solved it: the key follows the mechanism, not the runs. This is why ESM finds the batteries' half of the division of storage in Victoria et al. (2019), batteries for the day and hydrogen for weeks, but not the hydrogen half.

The other three levers fail on mechanism. With biomethane plentiful, a higher carbon price cannot reorder two fuels that are both carbon-free, (a). Without the industrial demand, (c), hydrogen gets no cheaper for the turbines: it is not short, and the electrolysers would run only to fill storage for the turbines' few hours, their capital spread over those hours (the Explorer's *Hydrogen as storage only* builds nothing of the chain). Without imports, (e), supply is taken away and hydrogen gets dearer.

2. Electrolysers as a flexible load

(i) The electrolyser's threshold. With a reformer burning biomethane at the margin, a MWh of hydrogen is worth the biomethane grossed up for the losses of reforming,

$$\lambda^{H_2} = \frac{p^F_{\text{bio}}}{\eta_{\text{SMR}}} = \frac{62}{0.72} \approx 86 \text{ €/MWh}.$$

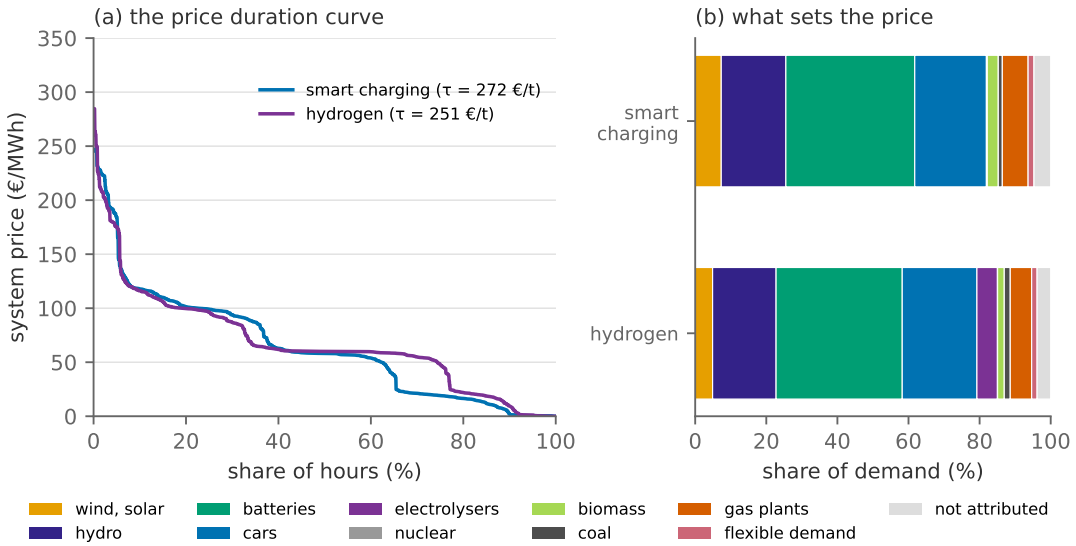


Figure S8: ESM, Fig. 8.9: smart charging and the hydrogen scenario at a cut of 99 %: (a) the system's price duration curve; (b) what sets the price, as shares of the year's demand.

An electrolyser turns a MWh of electricity into $\eta^{EL} = 0.7$ MWh of hydrogen, so by its condition (ESM, appendix E) it runs flat out whenever

$$\lambda_{z,h} < \eta^{EL}\lambda^{H_2} = 0.7 \times 86 \approx 60 \text{ €/MWh}.$$

Running part-loaded, it is the marginal consumer, and the price equals that value: the long step at about 60 €/MWh in the duration curve, which runs on well after the price without electrolysers has fallen away (Figure S8(a)). Running flat out in the cheapest hours, the electrolysers take wind and solar that would otherwise be curtailed: there are fewer hours near zero (782 against 930 below 5 €/MWh), and the share of wind and solar curtailed falls from 7 to 3 %. In the dear hours they step aside, so the expensive tail remains. A new load, however flexible, fills the cheap tail without emptying the dear one, the qualification of ESM §5.2 at scale, and the average rises (68 against 64 €/MWh). The two scenarios are not like for like: each budget is 1 % of its own scenario's emissions without one, so they differ in tonnes. In the Explorer: Hydrogen › *Electrolysers against the electricity price: DE, Germany*, and Prices and margins › *The price-duration curve*, the reference against *Smart charging*.

(ii) Key: (b). An electrolyser earns what its hydrogen is worth less what its electricity costs, $\eta^{EL}\lambda^{H_2}_{z,h} - \lambda_{z,h}$ per MWh it draws, and its multiplier on full output is $\overline{\mu}^{EL}_{z,h} = \max\left(\eta^{EL}\lambda^{H_2}_{z,h} - \lambda_{z,h},\, 0\right)$. Part-loaded, it is the marginal consumer and the price equals the value of its hydrogen, so it earns nothing towards its capital. Flat out, it buys electricity cheaper than that value, and the difference is a rent, as a plant's is when the price is above its running cost. Summed over the year, the rents just pay for it, $\sum_h \overline{\mu}^{EL}_{z,h} = F^{EL}$ (ESM, appendix E).